

A further conjecture on OEIS A208138, settled by the proof already in hand

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A208138 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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1 The premise

The entry's first terms are 9, 81, 324, 900, 2025, 3969, 7056, 11664, $\dots$, with offset 1. The sequence satisfies

$$a(n) = 5a(n-1) - 10a(n-2) + 10a(n-3) - 5a(n-4) + 1a(n-5),$$

a linear recurrence of order 5 with characteristic polynomial

$$q(x) = x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 1.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least $5 + 2$ further terms after that index.

2 The further conjecture

Separately from the conjecture already settled, the entry records

$$a(n) = 9 \cdot (n^2(1+n)^2) / 4.$$

The further claim is a CLOSED FORM, and for that the divisibility is not the whole argument. It gives that a and the closed form f satisfy the same linear recurrence; it does not say they are the same solution of it. Write p for the annihilator of f ,

$$p(x) = x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 1,$$

of order $D = 5$. Since $q \mid p$ and a satisfies the recurrence of q , a satisfies that of p as well; f satisfies it by construction. Two solutions of a linear recurrence of order D with nonzero leading coefficient agree everywhere as soon as they agree at D consecutive indices, and a and f were checked to agree at $n = 6, \dots, 10$, which lies inside the terms the entry publishes and past the index from which a provably satisfies p . Hence $a(n) = f(n)$ for every n in the claimed range.

Theorem 1. *The conjecture displayed above is true.*

3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly q says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.